Solutions, Eleventh Annual ECC Undergraduate Mathematics Competition, April 5, 2008

1. Adding logs.

It is $\boxed{p = \dfrac{q}{q-1}, \quad \text{for } q \neq 1}$. For,

$$\log(p+q) = \log p + \log q = \log pq \Rightarrow p + q = pq$$

(because the log function is strictly increasing, and therefore one-to-one). Solving for p we obtain

$$p = \frac{q}{q-1}.$$

The domain is $\boxed{\{q : q > 0,\ q \neq 1\}}$. We note that $q = 1$ is impossible in the original equation because that would require $\log p = \log(p+1)$. That $q > 0$ is specified in the problem statement.

2. Unknown coefficients.

$\boxed{A = 5 \text{ and } B = -4}$. We have

$$0 = P(1) = 2 + A + B - 3,$$

and

$$9 = P(-2) = -16 + 4A - 2B - 3,$$

so $A + B = 1$ and $4A - 2B = 28$. Solve simultaneously, to get $A = 5$ and $B = -4$.

3. A common root.

They are $\boxed{b = \pm 10010}$. If r is a common root, then the first equation implies that

$$b = \frac{-2008r^2 - 8002}{r}$$

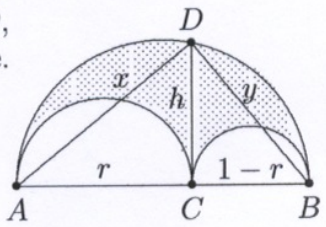
and the second implies that

$$b = \frac{-8002r^2 - 2008}{r}.$$

Therefore we must have $2008r^2 + 8002 = 8002r^2 + 2008$; i.e., $5994r^2 = 5994$. Thus, $r = \pm 1$. Corresponding to $r = 1$ we have $b = -10010$, and with $r = -1$, we have $b = 10010$.

4. Area of the arbelos.

Without loss of generality we take the diameter of the large semicircle to be 1. Let $r = AC$, and then $1 - r = CB$. Let $x = AD$, $y = BD$ and $h = CD$, and note that angle ADB is a right angle. Then



$$x^2 = h^2 + r^2$$
$$y^2 = h^2 + (1-r)^2$$
$$1^2 = x^2 + y^2$$

From the first two equations we have $x^2 - y^2 = 2r - 1$. This, together with the third equation, gives $2x^2 = 2r$, so $x^2 = r$, $y^2 = 1 - r$ and $h^2 = r - r^2$. The area of the arbelos is

$$\frac{\pi}{2}\left(\frac{1}{2}\right)^2 - \frac{\pi}{2}\left(\frac{r}{2}\right)^2 - \frac{\pi}{2}\left(\frac{1-r}{2}\right)^2 = \frac{\pi}{8}(1 - r^2 - 1 + 2r - r^2)$$
$$= \frac{\pi}{8}(2r - 2r^2) = \pi\left(\frac{h}{2}\right)^2,$$

which is the area of the circle with diameter h.

5. Solve for x.

The roots are $\boxed{x = 2008^{1\pm\sqrt{1/2}}}$. Let $u = \log_{2008} x$. Then $x = 2008^u$, and we may write the equation

$$(2008)^{1/2}(2008^u)^u = (2008^u)^2,$$

which implies that $u^2 + 1/2 = 2u$; i.e., $2u^2 - 4u + 1 = 0$. The solutions of this quadratic are $u = 1 \pm \sqrt{1/2}$, so we have

$$x = 2008^{1\pm\sqrt{1/2}}.$$

6. Probability of three heads.

The probability of exactly 3 heads in 5 tosses is $\boxed{\dfrac{40}{243}}$. Let p be the probability of heads on a single toss. Then, from what is given,

$$\binom{5}{1}p(1-p)^4 = \binom{5}{2}p^2(1-p)^3, \qquad \text{with} \quad p(1-p) \neq 0.$$

This equation simplifies to $1 - p = 2p$, from which we find that $p = \frac{1}{3}$. Then the probability of exactly 3 heads in 5 tosses is

$$\binom{5}{3}\left(\frac{1}{3}\right)^3\left(\frac{2}{3}\right)^2 = \frac{10 \cdot 4}{3^5} = \frac{40}{243}.$$

7. Limit of a function.

The limit is $\boxed{7/6}$. We have

$$\lim_{x\to-\infty} \frac{(3x+\sqrt{9x^2-7x+2})(3x-\sqrt{9x^2-7x+2})}{3x-\sqrt{9x^2-7x+2}} = \lim_{x\to-\infty} \frac{7x-2}{3x-\sqrt{9x^2-7x+2}}.$$

If we put $x=-y$ and changing sign in numerator and denominator, this becomes

$$\lim_{y\to\infty} \frac{7y+2}{3y+\sqrt{9y^2+7y+2}} = \lim_{y\to\infty} \frac{7+\frac{2}{y}}{3+\sqrt{9+\frac{7}{y}+\frac{2}{y^2}}} = \frac{7}{6}.$$

8. Minimum value.

The minimum value is $\boxed{10}$. By the AM-GM inequality,

$$\frac{x}{3y}+\frac{6y}{z}+\frac{4z}{x} \geq 3\sqrt[3]{\left(\frac{x}{3y}\right)\left(\frac{6y}{z}\right)\left(\frac{4z}{x}\right)} = 3\cdot 2 = 6.$$

This value 6 will be achieved when each of the factors under the radical is equal to 2:

$$2 = \frac{x}{3y} = \frac{6y}{z} = \frac{4z}{x};$$

i.e., for any values of x, y, z with $x = 6y = 2z$. (E.g., $x=6, y=1, z=3$.) Thus the minimum value of

$$\frac{x}{3y}+\frac{6y}{z}+\frac{4z}{x}+4$$

is $6+4=10$.

9. How many positive integer pairs?

There are $\boxed{21}$ such pairs. The equation is equivalent to $xy-2008x-2008y=0$; thus, to

$$(x-2008)(y-2008) = 2008^2.$$

In view of the prime factorization $2008^2 = 2^6(251)^2$, we see that the factor $x-2008$ must have the form $2^a(251)^b$, where a and b are integers, $0\leq a\leq 6$ and $0\leq b\leq 2$. There are $7\cdot 3 = 21$ such choices for (a,b), and thus 21 different possible values for $x-2008$, and for each of them, a single value for $y-2008$.

10. An integral.

The value is $\boxed{\frac{\pi}{12}}$. Call the value I. Using the substitution $y = \frac{\pi}{2} - x$ and the fact that $\cos x = \sin(\frac{\pi}{2} - x)$ we obtain

$$I = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sin^3 x dx}{\sin^3 x + \cos^3 x} = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sin^3 x dx}{\sin^3 x + \sin^3(\frac{\pi}{2} - x)} = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sin^3(\frac{\pi}{2} - y)dy}{\sin^3(\frac{\pi}{2} - y) + \sin^3 y}.$$

Adding the last two expressions for I we obtain

$$2I = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sin^3 x + \sin^3(\frac{\pi}{2} - x)}{\sin^3 x + \sin^3(\frac{\pi}{2} - x)} dx = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} 1 dx = \frac{\pi}{3} - \frac{\pi}{6} = \frac{\pi}{6},$$

and thus $I = \pi/12$.